

AI UX & Data Visualisation Design Principles (CA6002)

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1

1

Chapter 4.2 – Human and AI Centred

Contents

- Human + AI
- Human + AI – What AI is good at
- Human + AI – What humans do better
- Human + AI – The missing middle



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2

2

Human and AI Centred

Human + AI

- An often heard narrative is that **humans** and **AI** are **adversaries**, fighting for each other's job. Such talks detract from the fact that there are many opportunities for **positive collaborations** between the two.
- Humans and AI systems can be **symbiotic partners**, complementing and improving each other's capabilities towards better outcomes.
- In this symbiotic relationship, AI can give **humans improved** capabilities and services while the humans can help the **AI improve** its performance.
- E.g. the personalisation recommendation feature of the Netflix video streaming service **makes it easier** to find interesting movies to watch. The AI algorithm improves its accuracy through a symbiotic relationship with the users by **learning** their preferences and movie watching behaviour with usage.



- [1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022
- [2] Sammer Chhabara, MobileSyrup.com - <https://mobilesyrup.com/2017/08/22/80-percent-netflix-shows-discovered-recommendation/>

3

3

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Humans + AI - What AI is good at

- **Monotonous maestro** – AI systems are very good at performing routine and repetitive tasks. They do so consistently, fast and tirelessly.
- **Distributive spread** – they can be easily replicated and can share information through the network instantaneously, multiplying their efficacy (i.e. as one AI system learns, all other systems benefit).
- **Ubiquitous presence** – AI services can be accessed from any locality and on multiple hardware devices (unlike humans who cannot separate brain from physical body).
- **Big data** – they can analyse vast quantity of data and spot hidden patterns much faster than humans (e.g. AI systems can be used to make time-sensitive prediction and detect anomalies in data streams, spot hidden patterns in huge datasets).



- [1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022
- [3] IBM developers, What is anomaly detection? - <https://developer.ibm.com/learningpaths/get-started-anomaly-detection-api/what-is-anomaly-detection/>

4

4

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Humans + AI - What humans do better

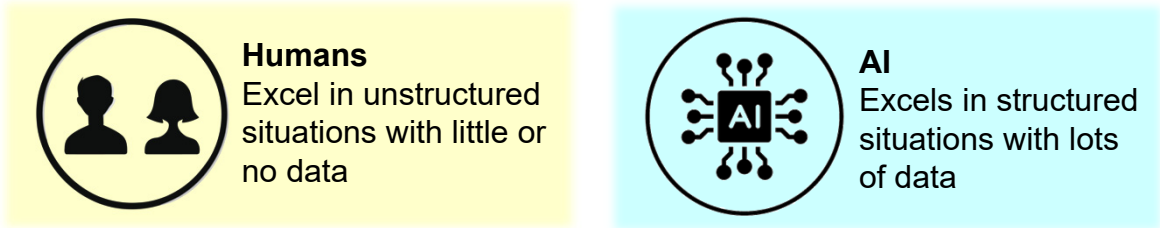
- **VUCA scenarios** – humans are good at **interpreting** and quickly **adapting** to situations and information that are volatile, uncertain, complex and ambiguous.
- **Physical dexterity** – humans have better agility and coordination to interact with the dynamic physical world and performing intricate operations (e.g. manipulating wires).
- **Valued judgement** – while AI is good at routine decision, human judgement is needed when outcomes involve ethical values, payoff vs utility considerations.
- **Small data** – humans can learn quickly with small amounts of data (e.g. a child quickly learns what a bicycle looks like with only a few exposure).
- **Social engagement** – humans have better interpersonal skills (i.e. they are better in interpreting social, emotional and non-verbal cues).

5

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Human + AI - The Missing Middle

- Daugherty & Wilson argues that companies achieve the largest performance boost when humans and AI (machines) **work together** as **allies** (i.e. partners).
- Humans and AI excel at different types of tasks and businesses require both kinds of capabilities, and it is in the **missing middle** where this type of **collaborative teamwork** occurs.



6

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Human + AI - The Missing Middle

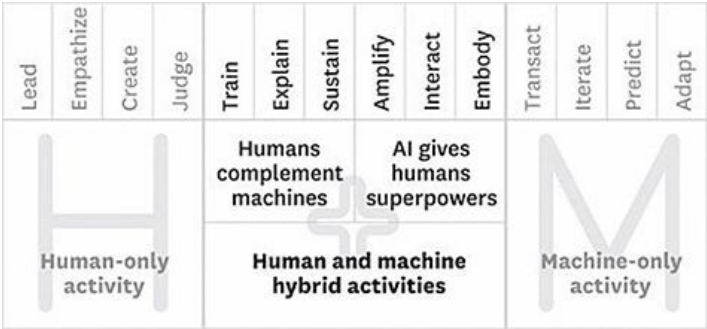
Role of Humans

- **Train** AI to perform task
- **Explain** outcome of AI
- **Sustain** AI in a responsible manner

Role of AI

- **Amplify** human insight, intuition & creativity
- **Interact** with humans at scale using novel interfaces
- **Embody** physical attributes that extend a person's capabilities

Six roles in the missing middle were identified.



7

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Human + AI - The Missing Middle

Trainers

Reinforcement Learning from Human Feedback (RLHF)

It's a classic growth flywheel that compounds over time for OpenAI.

Better Models

More Feedback

ChatGPT

More Users

Humans help reduce ChatGPT's propensity to 'hallucinate' and improve the model's answers using reinforcement learning from human feedback

Explainers

Explainable AI breaks down the risk model, providing a clear explanation for each decision.

Financial Data

Risk Model

Explainable AI

User

Critical use of AI need experts (e.g. algorithm forensic analyst) who can hold complex algorithms accountable for its decision by explaining its inner workings

Sustainers

practices

standards

strategy

policy

rules

control

requirements

regulations

audit

risks

COMPLIANCE

Ensuring the proper use of AI is the responsibility of a sustainer (e.g. ethics compliance manager) who continually ensures the AI system functions properly

8

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Human + AI - The Missing Middle

Amplify	Interaction	Embodiment
		
AI tools like smart glasses amplifies maintenance workers productivity by overlaying context sensitive information in their field of view while they work	Agent Assist uses NLP to understand queries, retrieves relevant answers from FAQ databases to provide instant help to customer care reps.	A co-bot works closely with a human worker to ensure perfect fit and with the co-bot he avoids heavy lifting and having to access hard-to-reach places

9

Human and AI Centred

Summary

- Humans and AI systems have their respective strengths and weaknesses. Humans excel at unstructured tasks that require **value** judgement and **social** engagement. Whole, AI excels in **distributed** tasks that are high **repetitive** and involve large amounts of **data**.
- Daugherty & Wilson coined the phrase “**missing middle**” to describe the **sybiotic space** where humans and AI collaborate to yield the best outcome.
- Instead of focusing on AI purely as a replacement for human labour, they argued that the greatest value comes from **augmenting** human capabilities with machines.
- The "missing middle" is this crucial but often overlooked collaboration area that exists between entirely human-only and entirely machine-only tasks.

10

References for Chapter 4.2 – Human and AI Centred

[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

[2] Sammer Chhabara, MobileSyrup.com - <https://mobilesyrup.com/2017/08/22/80-percent-netflix-shows-discovered-recommendation/>

[3] IBM developers, What is anomaly detection? - <https://developer.ibm.com/learningpaths/get-started-anomaly-detection-api/what-is-anomaly-detection/>

[4] P.R. Daugherty, H.J. Wilson, Human + Machine: Reimagining Work in the Age of AI, Harvard Business Review Press, 2024

[5] Venture Beat, How Stitch Fix used AI to personalize its online shopping experience, Jul 2020 - <https://venturebeat.com/business/how-stitch-fix-used-ai-to-personalize-its-online-shopping-experience/>



Note: All online articles were accessed and available on 12 Dec 2025



Think and Apply

AI Products that are Human and AI Centred

- How are the legal and creative sectors impacted by Generative AI technologies?
- How does the design of these AI-powered products seeks to harness the symbiotic partnership between humans and AI?
- Robotic surgical systems have high automation and high human control. Yet their pervasive use is giving rise to unintended consequences. What might these be?





12